

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 21/01/2026**
Time: 3 Hrs **GTM - 21** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Assertion (A): Net dipole moment of a polar linear isotropic dielectric substance is non-zero even in the absence of an external electric field.

Reason (R): In the absence of an external electric field, the different permanent dipoles of a polar dielectric substance are oriented in random directions.

In the light of the above statements, choose the most appropriate answer from the options given below:

A) A is correct but R is not correct.

B) Both A and R are correct but R is not the correct explanation of A.

C) Both A and R are correct and R is the correct explanation of A.

D) A is not correct but R is correct.

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2. In a moving coil galvanometer, two moving coils M_1 and M_2 have the following particulars:

$R_1 = 5\Omega,\ N_1 = 15,\ A_1 = 3.6\times10^{-3}\ m^2,\ B_1 = 0.25\ T$

$R_2 = 7\ \Omega,\ N_2 = 21,\ A_2 = 1.8\times10^{-3}\ m^2,\ B_2 = 0.50\ T$

Assuming that torsional constant of the springs are same for both coils, what will be the ratio of current sensitivity of M_1 and M_2 ?

A) 1 : 1 B) 1 : 4 C) 5 : 7 D) 1 : 2

3. The moment of inertia of a circular ring of mass M and diameter r about a tangential axis lying perpendicular the plane of the ring is

A) $\frac{1}{2}Mr^2$ B) $\frac{3}{8}Mr^2$ C) $\frac{3}{2}Mr^2$ D) $2\,Mr^2$

4. If a drop of liquid of surface tension T breaks into smaller droplets, it results in lowering of temperature of the droplets. Let a drop of radius R, break into N small droplets each of radius r. Estimate the drop in the temperature. (ρ density of liquid, s-specific heat of liquid)

A) $\frac{3T}{2\rho s}\left[\frac{1}{R}-\frac{1}{r}\right]$ B) $\frac{3T}{\rho s}\left[\frac{1}{R}-\frac{1}{r}\right]$ C) $\frac{6T}{\rho s}\left[\frac{1}{R}-\frac{1}{r}\right]$ D) $\frac{3T}{\rho s}\left[\frac{1}{R}+\frac{1}{r}\right]$

5. An electron (mass m) with an initial velocity $v = v_0 i (v_0 > 0)$ is in an electric field $E = -E_0\hat{i}$ (E_0 = constant > 0). It's de-Broglie wavelength at time t is given by

A) $\frac{\lambda_0}{\left(1-\frac{eE_0t}{mv_0}\right)}$ B) $\frac{\lambda_0}{\left(1+\frac{eE_0t}{mv_0}\right)}$ C) λ_0 D) $\lambda_0 t$

6. Equation of a plane progressive wave is given by $y = 0.6\sin 2\pi\left(1-\frac{x}{2}\right)$. On reflection from a denser medium its amplitude becomes $\frac{2}{3}$ of the amplitude of the incident wave. The equation of the reflected wave is

A) $y = 0.6\sin 2\pi\left(t+\frac{x}{2}\right)$ B) $y = -0.4\sin 2\pi\left(t+\frac{x}{2}\right)$

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C) $y = 0.4\sin 2\pi\left(t + \frac{x}{2}\right)$ D) $y = -0.4\sin 2\pi\left(t - \frac{x}{2}\right)$

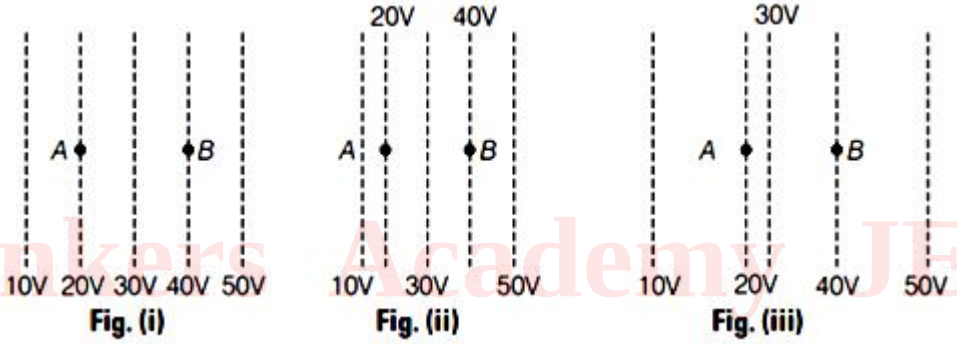
7. If momentum (p), area (A) and time (T) are taken to be fundamental quantities, then energy has the dimensional formula.

A) $\left[pA^{-1}T^{1}\right]$ B) $\left[p^{2}AT\right]$ C) $\left[p\,A^{-1/2}\,T\right]$ D) $\left[pA^{1/2}\,T^{-1}\right]$

8. A solenoid having area A and length 'ℓ' is filled with a material having relative permeability 0.5. The magnetic energy stored in the solenoid is

A) $\frac{B^2 A\ell}{\mu_0}$ B) $\frac{B^2 A\ell}{2\mu_0}$ C) $B^2\,A\ell$ D) $\frac{B^2 A\ell}{4\mu_0}$

9. Figure shows some equipotential lines distributed in space. A charged object is moved from point A to point B.



A) The work done in fig. (i) is the greatest

B) The work done in fig. (ii) is least

C) The work done is the same in fig. (i), fig. (ii) and fig. (iii)

D) The work done in fig. (iii) is greater than fig. (ii) but equal to that in

10. Which of the processes described below are reversible ?

A) The increase in temperature of an iron rod by hammering it.

B) A gas in a small container at a temperature T_1 is brought in contact with a big reservoir at a higher temperature T_2 which increases the temperature of the gas.

C) A quasi-static isothermal expansion of an ideal gas in cylinder fitted with a frictionless piston.

D) An ideal gas is enclosed in a piston cylinder arrangement with adiabatic walls. A weight w is added to the piston, resulting in compression of gas.

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11. Taking the Bohr radius as $a_0 = 53\ pm$, the radius of Li^{++} ion in its ground state, on the basis of Bohr's model, will be about (nearly)

A) 53 pm B) 27 pm C) 18 pm D) 13 pm

12. M_x and M_y denote the atomic masses of the parent and the daughter nuclei respectively in radioactive decay. The Q-value for a β^- decay is Q_1 and that for a β^+ decay is Q_2. If m_e denotes the mass of an electron, then which of the following statements is correct ? (C $\rightarrow$ speed of light)

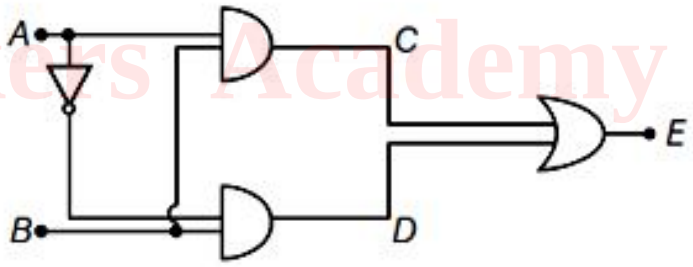
A) $Q_1 = (M_x - M_y)C^2$ *and* $Q_2 = \left[M_x - M_y - 2m_e\right]C^2$

B) $Q_1 = (M_x - M_y)C^2$ *and* $Q_2 = (M_x - M_y)C^2$

C) $Q_1 = (M_x - M_y - 2m_e)C^2$ *and* $Q_2 = (M_x - M_y + 2C_e)C^2$

D) $Q_1 = (M_x - M_y + 2m_e)C^2$ *and* $Q_2 = (M_x - m_y + 2m_e)C^2$

13. Truth table for the given circuit is



(a)

A	B	E
0	0	1
0	1	0
1	0	1
1	1	0

(b)

A	B	E
0	0	1
0	1	0
1	0	0
1	1	0

(c)

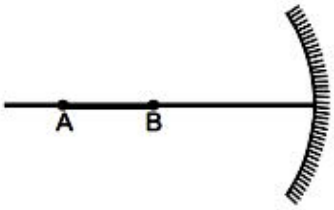
A	B	E
0	0	0
0	1	1
1	0	0
1	1	1

(d)

A	B	E
0	0	0
0	1	1
1	0	1
1	1	0

14. A linear object AB is placed along the axis of a concave mirror. The object is moving towards the mirror with speed v. The speed of the image of the point A is 4v and the speed of the image of B is also 4v. If centre of the line AB is at a distance L from the mirror, then length of the object AB will be

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A) $\frac{3L}{2}$ B) $\frac{5L}{3}$ C) L D) $\frac{4L}{3}$

15. For a particle performing uniform circular motion, choose the incorrect statement from the following.

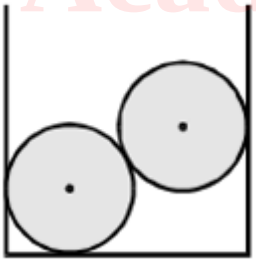
A) Magnitude of particle velocity remains constant

B) Particle velocity remains directed perpendicular to radius vector

C) Direction of acceleration keeps changing as particle moves

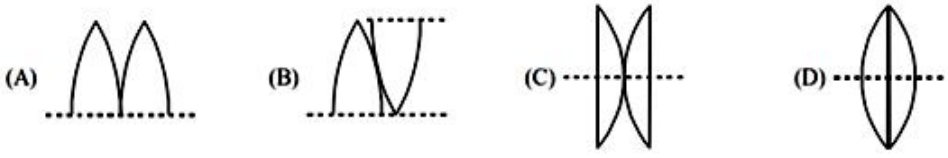
D) Angular momentum is constant in magnitude but direction keeps changing

16. Two smooth spheres each of radius 5 cm and weight W rest one on the other inside a fixed smooth cylinder of radius 8 cm. The reactions between the spheres and the vertical side of the cylinder are:



A) $\frac{W}{4}$ *and* $\frac{3W}{4}$ B) $\frac{W}{4}$ *and* $\frac{W}{4}$ C) $\frac{3W}{4}$ *and* $\frac{3W}{4}$ D) W and W

17. A convex lens is cut into two parts in different ways that are arranged in four manners as shown. Which arrangement will give maximum optical power ?



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18. A screw gauge with a pitch of 0.5mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5mm and the 25th division coincides with the main scale line?

A) 0.70 mm B) 0.50 mm C) 0.75 mm D) 0.80 mm

19. Match List-I with List-II.

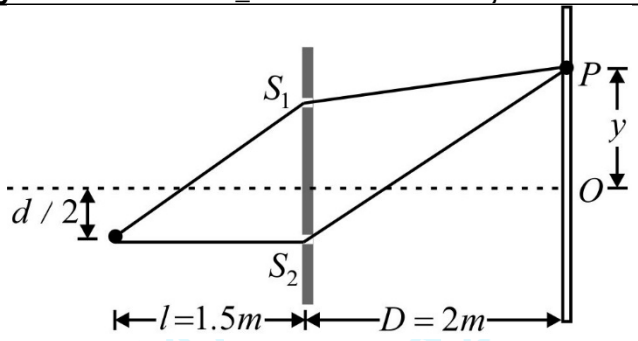
List-I		List-II	
A)	Coefficient of viscosity	I)	$\left[M\ L^2T^{-2}\right]$
B)	Surface Tension	II)	$\left[M\ L^2T^{-1}\right]$
C)	Angular momentum	III)	$\left[M\ L^{-1}T^{-1}\right]$
D)	Rotational kinetic energy	IV)	$\left[M\ L^0T^{-2}\right]$

A) A-II, B-I, C-IV, D-III B) A-I, B-II, C-III, D-IV

C) A-III, B-IV, C-II, D-I D) A-IV, B-III, C-II, D-I

20. In the YDSE, the monochromatic source of light of wavelength λ is placed at a distance $\frac{d}{2}$ from the central axis (as shown in the figure), where d is the separation between the two slits S_1 and S_2. Find the minimum thickness of the film of refractive index $\mu = 1.5$ to be placed infront of S_2 so that intensity at O becomes $\frac{3}{4}th$ of the maximum intensity. Given $\lambda = 6000\,A^0$ and $d = 6mm$.

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A) $2000\,A^0$ B) $2500\,A^0$ C) $3000\,A^0$ D) $6000\,A^0$

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A string is wrapped around the rim of a wheel of moment of inertia $0.40\ kgm^2$ and radius 10 cm. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a force of 40 N. The angular velocity of the wheel after 10 s is x rad/s, where x is

22. The internal energy of air in $2m \times 4m \times 3m$ sized room at 1 atmospheric pressure will be _____ $\times 10^6\,J$ (nearly). (Consider air as diatomic molecule)

23. The refractive index of a transparent liquid filled in an equilateral hollow prism is $\sqrt{2}$. The angle of minimum deviation for the liquid will be _____ (in degrees).

24. The length of a light string is 1.4 m when the tension in it is 5N. If the tension increases to 7N, the length of the string is 1.56m. The original length of the string is _____ m.

25. An asteroid is moving directly towards the centre of the earth. When at a distance of 10R (R is the radius of the earth) from the earths centre, it has a speed of 12 km/s. Neglecting the effect of earth's atmosphere, what will be the speed of the asteroid when it hits the surface of the earth (escape velocity from the earth is 11.2 km/s) ? Give your answer to the nearest integer in kilometer/s_____.

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. A pentapeptide, 'α' on complete hydrolysis produced glycine (Gly), alanine (Ala), valine (val), Leucine (Leu), isoleucine (Ile) in equimolar proportion each. The number of pentapeptides (sequences) possible involving each of these amino acids is

A) 24 B) 32 C) 120 D) 130

27. Statement-(I): Isopentane contains nine primary hydrogen atoms, two secondary hydrogen atoms and one tertiary hydrogen atom.

Statement-(II): Boiling point of isopentane is higher than n-pentane.

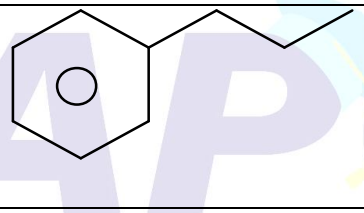
A) Statement-I is incorrect but Statement-II is correct.

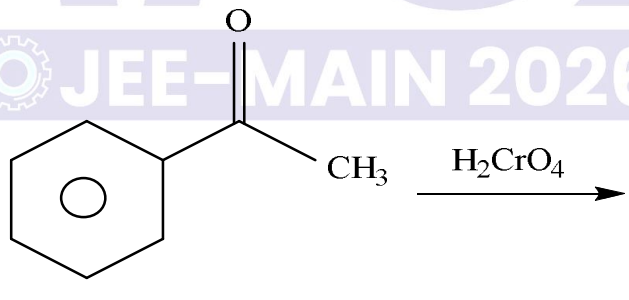
B) Statement-I is correct but Statement-II is incorrect.

C) Both Statement-I and Statement-II are correct.

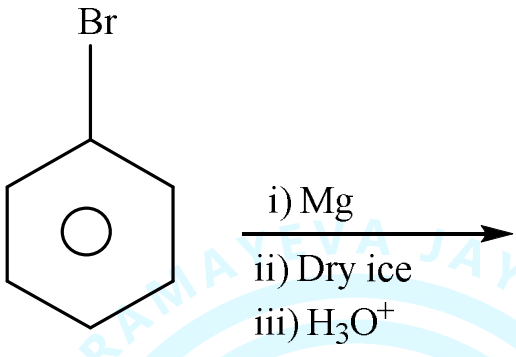
D) Both Statement-I and Statement-II are incorrect.

28. Consider the following reactions. From these reactions which reaction will not give carboxylic acid as major product ?

A) 

B) 

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C)	
D)	$R - CN \xrightarrow[ii)\ H_3O^+]{i)\ SnCl_2 + HCl}$

29. The nature of hydride (H_2S) and oxide (SO_2) formed by S, respectively are

A) basic and oxidizing B) basic and reducing

C) acidic and reducing D) acidic and oxidizing

30. In Duma's method for estimation of nitrogen, 0.3 g of an organic compound gave 50 mL of nitrogen collected at 300 K temperature and 715 mm pressure. The percentage composition of nitrogen in the compound (Aqueous tension at 300 k = 15 mm Hg) is

A) 18.67 B) 12.57 C) 20.87 D) 17.46

31. In 3,4-dibromo-5-phenylpentyne, there are …….. sp^3, …… sp^2 and …… sp hybridized carbon atoms respectively.

A) 3, 6, 2 B) 3, 8, 2 C) 4, 6, 2 D) 9, 6, 2

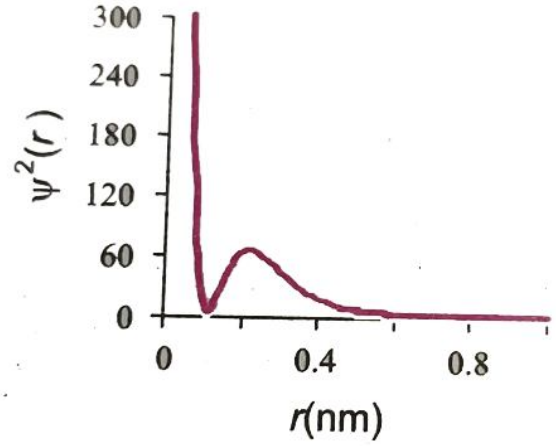
32. Which of the following statements are true ?

I) The magnetic orbital quantum number (m_ℓ) designates the orientation of the orbital.

II) Energy of 2s orbital of hydrogen atom is greater than that of 2s orbital of lithium.

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III)



is the variation of probability density $\Psi^2_{(r)}$ of 2s orbital as a function of distance (r) of the electron from the nucleus.

IV) The wave function of $2P_x$ orbital is zero every where in the xy plane.

Choose the correct answer from the options given below.

A) III and IV only

B) I and III only

C) I, II and III only

D) I, II and IV only

33. For the reaction $2A_{(g)} + B_{(g)} \rightarrow 2D_{(g)}$

$\Delta U^0 = -10.5kJ$ and $\Delta S^0 = -44.1\ JK^{-1}$

The ΔG^0 value in Joules at 298K for the above reaction and the nature of process are (R=8.3 J/K/mol)

A) 168 and nonspontaneous

B) 0.168 and nonspontaneous

C) -236 and spontaneous

D) -0.236 and spontaneous

34. Which of the following statement is incorrect

A) In nucleic acids, nucleotides are joined together by phosphodiester linkage

B) The bases present in DNA are adenine guanine, cytocine and uracil

C) Complete hydrolysis of DNA or RNA yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compounds.

D) In DNA molecules, the sugar moiety is β – D – 2 – deoxyribose

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35. The crystal field stabilization energy (CFSE) of $[Fe(H_2O)_6]Cl_2$ and $K_2[NiCl_4]$ respectively are

A) $-0.4\,\Delta_0$ *and* $-0.8\,\Delta t$ B) $-0.4\,\Delta_0$ *and* $-1.2\,\Delta t$

C) $-2.4\,\Delta_0$ *and* $-1.2\,\Delta t$ D) $-0.6\,\Delta_0$ *and* $-0.8\,\Delta t$

36. The type of hybridization and the magnetic property of $[Co(Ox)_3]^{-3}$ are

A) sp^3d^2 paramagnetic with four unpaired electrons

B) d^2sp^3 paramagnetic with four unpaired electrons

C) d^2sp^3 and diamagnetic D) sp^3d^2 and diamagnetic

37. Match List-I with List-II.

List-I (Purification technique)		**List-II (Mixture of organic compound)**	
A)	Simple Distillation	I)	Diesel + Petrol
B)	Fractional distillation	II)	Aniline + Water
C)	Distillation under reduced pressure	III)	$CHCl_3 + Ph-NH_2$
D)	Steam distillation	IV)	Glycerol + Spent lye

Choose the correct answer from the options given below:

A) A-II, B-III, C-IV, D-I B) A-II, B-IV, C-I, D-III

C) A-III, B-IV, C-II, D-I D) A-III, B-I, C-IV, D-II

38. Which among the following molecules is (a) involved in sp^3d^2 hybridization (b) has same bond lengths and (c) has at least one lone pair of electrons on the central atom ?

A) SF_4 B) SF_6 C) $XeOF_4$ D) XeF_4

39. Formation of $Fe_4[Fe(CN)_6]_3$, a Prussian blue coloured complex formed by the addition of sodium hexacyanoferrate (II) in sodium carbonate extract of salt, indicates the presence of

A) Azide ion B) Cyanide ion C) Nitride ion D) Sulphide ion

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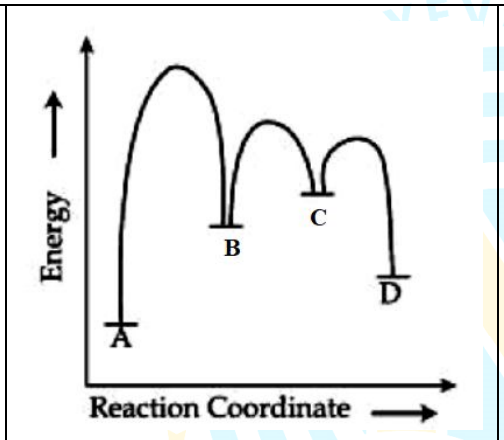
40. Reactant A converts to product D through the given mechanism (with the net absorption of heat):

$A \rightarrow B$; *slow* ; $\Delta H = +ve$

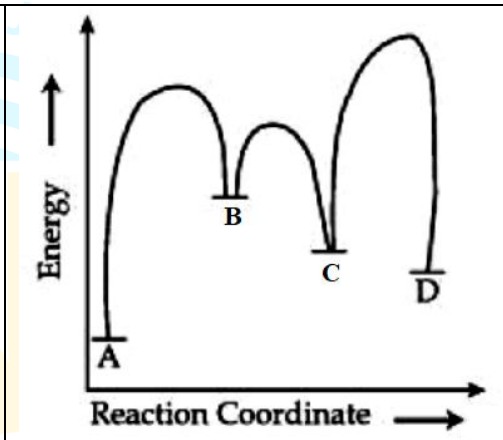
$B \rightarrow C$; *fast* ; $\Delta H = +ve$

$C \rightarrow D$; *fast* ; $\Delta H = -ve$

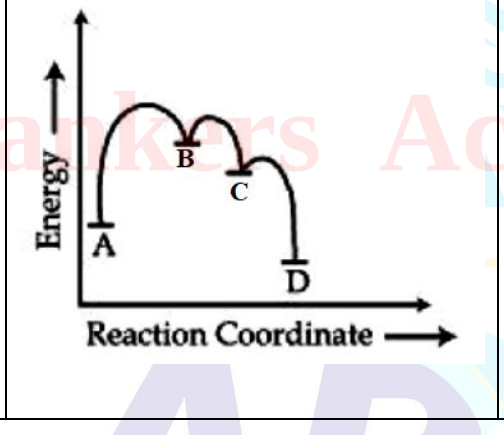
A)



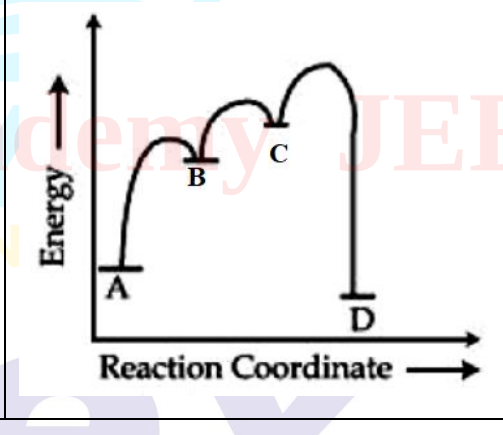
B)



C)



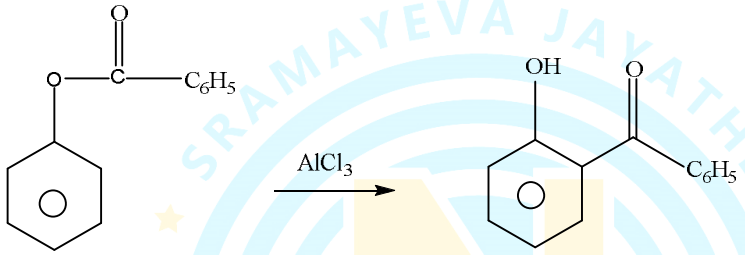
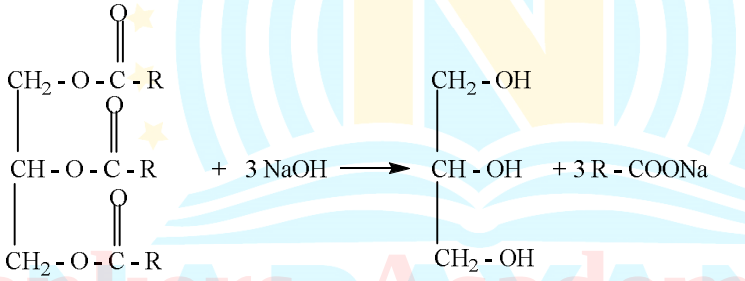
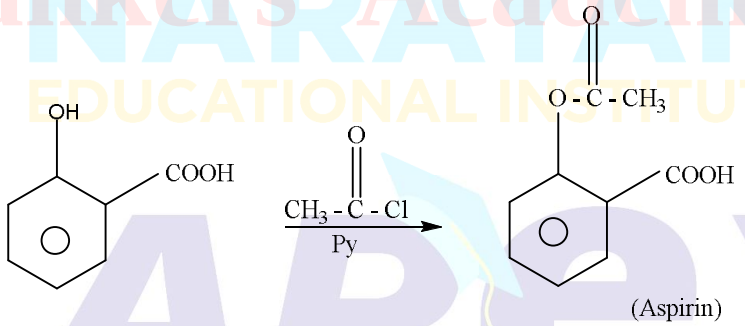
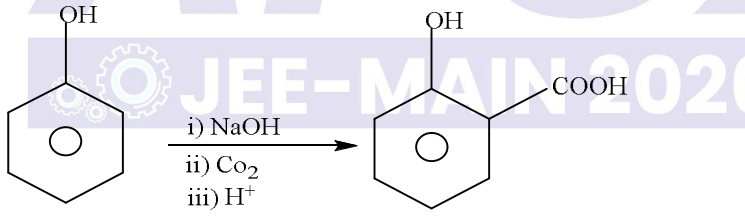
D)



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41. Match List-I with List-II.

List-I (Reaction)		List-II (Name of reaction)	
A)		I)	Saponification
B)		II)	Acetylation
C)		III)	Kolbe's reaction
D)		IV)	Molecular rearrangement

A) A – IV, B – II, C – I, D - III

B) A – IV, B – I, C – II, D - III

C) A – I, B – IV, C – II, D - III

D) A – IV, B – I, C – III, D - II

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42. Consider the following chemical equilibrium of the gas phase reaction at a constant temperature:

$A(g) \rightleftharpoons B(g) + C(g)$

If p being the total pressure, K_p is the equilibrium constant and α is the degree of dissociation, then which of the following is true at equilibrium ?

A) If p value is extremely high compared to K_p, $\alpha = 1$.

B) When p increases α decreases.

C) If K_p value is extremely high compared to p, α becomes much less than unity.

D) When p increases α increases.

43. Bromine monochloride, $BrCl$ decomposes into $'Br'$ and $'Cl'$ and reaches the equilibrium. $2BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}$ for which $K_C = 32$ at 500K. If initially pure $BrCl$ is present, at a concentration of $3.3\times10^{-3}\ mol/lit$. The molar concentration of $BrCl$ in the mixture at equilibrium is___ (in moles/lit)

A) 6×10^{-4} B) 3×10^{-4} C) 3.3×10^{-2} D) 6×10^{-2}

44. The pH of 0.1M solution of cyanic acid $(HCNO)$ is 2.3. The ionisation constant of the acid is

A) 5×10^{-3} B) 2×10^{-5} C) 2.5×10^{-4} D) 2.5×10^{-1}

45. Electronic configurations of four elements A, B, C and D are given below:

A) $[He]\,2s^2\,2p^3$ B) $[He]\,2s^2\,2p^4$

C) $[Ne]\,3s^2\,3p^3$ D) $[Ne]\,3s^2\,3p^4$

Which of the following is the correct order of increasing electronegativity (Pauling's scale) ?

A) C < B < A < D B) D < C < A < B

C) A < D < C < B D) C < D < A < B

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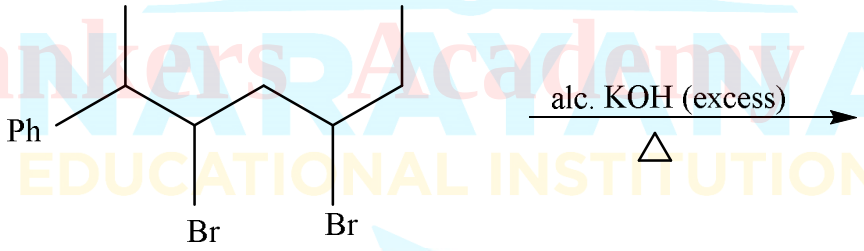
SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. The spin only magnetic moment value of M^{+2} ion in 3d series elements which is having, highest melting point is _____ (in nearest integer)

47. Two compounds of A and B elements are AB_2 and AB_4 dissolved in 20 gr of benzene. 1 gr AB_2 lowers the freezing point by $2.5K$ where as 1 gr of AB_4 lowers it by $2.0K$; K_f of benzene is $5.0K\,kg.mol^{-1}$. The atomic masses of A and B are $'x'$ and $'y'$, the value of $\frac{x}{y}$ is___

48. The number of sigma bonds present in the major product obtained in the following reaction is



49. A solution of 0.1 M CH_3COOH is placed between parallel electrodes of cross-section area $4\,cm^2$, separated by 2 cm. For this solution resistance measured is $100\,\Omega$. Calculate elevation in boiling point of the 0.1 M CH_3COOH solution using following informations.

Given: $K_b\,(H_2O) = 0.8\,k.\,kg\,.\,mol^{-1}$

$\Lambda_M^{\infty}\,(H^+) = 300\ s.\,cm^2\,.\,mol^{-1}$

$\Lambda_M^{\infty}\,(CH_3COO^-) = 100\ s.\,cm^2\,.\,mol^{-1}$

[Give your answer multiplying by 100].

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50. The solubility product constant of Ag_2CrO_4 and $AgBr$ are 3.2×10^{-11} and 2.5×10^{-11}. The ratio of molarities of Ag_2CrO_4 and $AgBr$ are $'x'$ and $'y'$. The value of $\frac{x}{y}$ is____

MATHEMATICS MAX.MARKS: 100

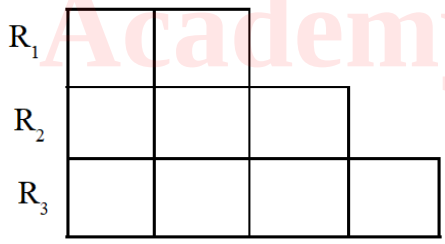
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let A, B be non-zero square matrices satisfying $A+BA^T=I$ and $B+AB^T=I$ and $A^4-2A^2=kA$ then k =

A) 1 B) -1 C) 0 D) 2

52. Number of ways six A's one in 1 small square can be placed in the figure given below so that no row remains empty is _____



A) 72 B) 6 C) 76 D) 66

53. In an increasing G.P. sum of first and last term is 66, the product of 2nd and second from the last term is 128. Sum of all terms is 126. Then number of terms in the progression is _____

A) 4 B) 6 C) 8 D) 12

54. Let $f(x)=\int_3^x \frac{dt}{\sqrt{t^4+3t^2+13}}$. If $g(x)=f^{-1}(x)$ then $g'(0)=$ _____

A) $\frac{1}{11}$ B) 11 C) $\sqrt{13}$ D) $\frac{1}{\sqrt{13}}$

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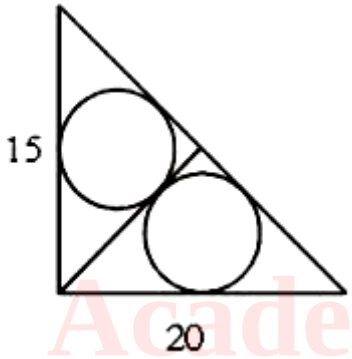
55. If $3\int_{-3}^{1}(x-k)(x+3)(x-1)\,dx = 64$ then k = _____

A) 1 B) -1 C) 2 D) 0

56. A parabola touches x-axis at $A(1,0)$ and y-axis at $B(0,1)$. If focus of the parabola divides AB in the ratio $m:n$ internally then m^2+n^2 = _____

A) 5 B) 13 C) 2 D) 3

57. In the right angle triangle as shown, an altitude is drawn from the right angle to the hypotenuse. Circles are inscribed within each of the smaller triangles. Then the distance between the centres of these circles is___



A) $5\sqrt{3}$ B) $7\sqrt{3}$ C) $8\sqrt{2}$ D) $\sqrt{50}$

58. An ellipse whose foci at $S(2,4), S'(14,9)$ touches x-axis then eccentricity of the ellipse _____

A) $\frac{12}{\sqrt{313}}$ B) $\frac{13}{\sqrt{313}}$ C) $\frac{14}{\sqrt{313}}$ D) $\frac{15}{\sqrt{313}}$

59. If $2\sec^2\alpha-\sec^4\alpha-2\operatorname{cosec}^2\alpha+\operatorname{cosec}^4\alpha=\frac{15}{4}$, then the value of $\tan\alpha$ is____

A) $\frac{1}{\sqrt{2}}$ B) $\frac{1}{2}$ C) $\frac{1}{2\sqrt{2}}$ D) $\frac{1}{4}$

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60. If the system of linear equations

$2x+3y-z=-2,\ x+y+z=4,\ x-y+|\lambda|z=4\lambda-4$, where $\lambda \in R$ has no solution then

A) $\lambda=9$ B) $\lambda=-7$ C) $\lambda=8$ D) $\lambda^2=1$

61. a_n, b_n be sequences of real numbers such that for all integers $n \geq 0$, $(3+i)^n = a_n + i\,b_n$ where $i=\sqrt{-1}$ then $\sum_{n=0}^{\infty}\left(\frac{a_n\,b_n}{12^n}\right)=$ _____

A) $\frac{9}{13}$ B) $\frac{5}{13}$ C) $\frac{12}{13}$ D) $\frac{7}{13}$

62. Out of 11 persons sitting around a roundtable, 3 persons, A, B, C are chosen at random the probability that no two of these are sitting next to each other is_____

A) $\frac{2}{15}$ B) $\frac{7}{11}$ C) $\frac{7}{15}$ D) $\frac{4}{9}$

63. If $L=\lim_{x\to 4}\left(\frac{\log_5\left(\frac{5x}{4}\right)}{\log_5\left(\frac{20}{x}\right)}\right)^{\frac{1}{\left(\log_5\left(\frac{x}{4}\right)\right)}}$ then $[L]=$ ___, where [.] is greatest integer function.

A) 6 B) 7 C) 8 D) 1

64. For each +ve integer n define $f_n(x)=\min\left\{\frac{x^n}{\lfloor n}, \frac{(1-x)^n}{\lfloor n}\right\}$ for $0 \leq x \leq 1$. Let $I_n=\int_0^1 f_n(x)\,dx$ for $n \geq 1$ then $\sum_{n=1}^{\infty}(I_n)=$ _____

A) $2\sqrt{e}-3$ B) $2\sqrt{e}-2$ C) $2\sqrt{e}-1$ D) $2\sqrt{e}$

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65. Coeff. of x^{15} in $\left(1+x+x^3+x^4\right)^n$ is _____

A) $\sum_{r=0}^{5} {}^nC_{15-3r} \cdot \left({}^nC_r\right)$ B) $\sum_{r=0}^{5} {}^nC_{5r}$

C) $\sum_{r=0}^{5} {}^nC_{3r}$ D) $\sum_{r=0}^{3} \left({}^nC_{3-r} \cdot {}^nC_{5r}\right)$

66. O is origin. Let PQR be arbitrary triangle. The point S is such that $\overrightarrow{OP}.\overrightarrow{OQ}+\overrightarrow{OR}.\overrightarrow{OS}=\overrightarrow{OR}.\overrightarrow{OP}+\overrightarrow{OQ}.\overrightarrow{OS}=\overrightarrow{OQ}.\overrightarrow{OR}+\overrightarrow{OP}.\overrightarrow{OS}$. Then S is

A) Centroid of ΔPQR B) Orthocentre of ΔPQR

C) Incentre of ΔPQR D) Circumcentre of ΔPQR

67. A line passing through origin cuts the lines $\frac{x+1}{1}=\frac{y-2}{1}=\frac{z-1}{-2}$ and $\frac{x-1}{1}=\frac{3y-8}{6}=\frac{z+3}{-1}$ at A and B. Then AB^2 = _____

A) 3 B) 6 C) 9 D) 12

68. $A=(1,2,3), B=(\alpha,4,6), C=(1,2,\beta)$ are three position vectors of ΔABC. If median AD is equally inclined with positive x, y, z axes then $\alpha+\beta=$____

A) 3 B) 5 C) 2 D) 4

69. The value of θ lying between 0 and $\frac{\pi}{2}$ satisfying the equation

$$\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix}=0$$ is _____

A) $\frac{5\pi}{24}, \frac{3\pi}{24}$ B) $\frac{7\pi}{24}, \frac{5\pi}{24}$ C) $\frac{7\pi}{24}, \frac{11\pi}{24}$ D) $\frac{\pi}{24}, \frac{11\pi}{24}$

70. Let $f(x)=\sum_{k=1}^{9}(2k-1)x^k; x\in R$. If $f'(2)+2f(2)=(2n+1)\times 2191$ then $n=$_____

A) 21 B) 23 C) 33 D) 17

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71. Let $\alpha,\beta \in R$, the mean and the variance of 6 observations $-3, 4, 7, -6, \alpha, \beta$ be 2 and 23 respectively. If the mean deviation about the mean of these 6 observations is $\frac{m}{n}(m,n$ are coprimes$)$, then $m+n$ is____

72. Number of integral values of m less than 50 so that the roots of quadratic equation $mx^2+(2m-1)x+m-2=0$ are rational is _____

73. The solution of $\left(2y+xy^3\right)dx+\left(x+x^2y^2\right)dy=0$ is $x^{\alpha}.y^{\beta}+\frac{x^3y^3}{3}=c$ then $\alpha+\beta=$_____

74. $x=\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+\sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}}+.....+\sqrt{1+\frac{1}{(2009)^2}+\frac{1}{(2010)^2}}$, then $\frac{(2010)x-2009}{2010}=\lambda$ then units digit of λ is _____

75. $4\cos\left(\cos^{-1}\left(\frac{1}{4}\left(\sqrt{6}-\sqrt{2}\right)\right)-\cos^{-1}\left(\frac{1}{4}\left(\sqrt{6}+\sqrt{2}\right)\right)\right)=$ _____

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